

Seok-Jin Kang

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Education

Korea University MS and Ph.D. in Biotechnology	Mar 2018 – Feb 2025
Korea University BS in Biotechnology	Mar 2014 – Feb 2018

Experience

Postdoctoral Fellow, Korea University Mar 2025 – Present
Immune Modulation Laboratory (PI: Taehoon Chun)

- Leading computational and experimental projects integrating **multi-omics** data with **machine learning and quantum-inspired models** to predict immunogenicity and design immune-specific biosensors.
- Applying **protein design frameworks** (AlphaFold, ProteinMPNN, and Rosetta) to engineer TCR–peptide–MHC interfaces for enhanced CAR-T specificity.
- Developing AI-driven pipelines for **CRISPR-based gene editing prediction** and analyzing **iPSC-derived T cell differentiation** using single-cell transcriptomics.

Laboratory Intern, Korea University Jan 2016 – Feb 2018
Immune Modulation Laboratory (PI: Taehoon Chun)

- Conducted molecular cloning, PCR/qPCR, and **gene expression analyses** to investigate host–virus interactions in porcine circovirus.
- Assisted in **cell-based immunoassays** (ELISA, flow cytometry) for evaluating immune responses to probiotic and viral vector candidates.
- Supported in vivo experiments involving **mouse immune challenge models**, including tissue collection and cytokine quantification.

Publications

*Note: Underlined names indicate **first authorship**.*

- JH Yoon, GB Yeon, H Lee, H An, J Oh, SJ Kang, IB Park, SA Lim, et al., **Tumor-targeted delivery of CXCL10 by mesenchymal stromal cells potentiates adoptive T cell therapy to treat solid tumors**, *Biomedicine & Pharmacotherapy* 192, 118579, 2025
- SJ Kang, H Shin, **Biophysical mechanisms of spider-silk constituting element–induced stick-slip behavior and hydrogen bond regeneration for high toughness in silk fibers**, *International Journal of Biological Macromolecules*, 147027, 2025
- SJ Kang, H Shin, **Amino acid sequence-based IDR classification using ensemble machine learning and quantum neural networks**, *Computational Biology and Chemistry*, 108480, 2025
- CH Lee, HJ Lee, SW Park, J Shin, SJ Kang, IB Park, HK Kim, T Chun, **Mutational analysis of pig tissue factor pathway inhibitor α to increase anti-coagulation activity in pig-to-human xenotransplantation**, *Biotechnology Letters* 46 (4), 521-530, 2024
- SW Park, IB Park, SJ Kang, J Bae, T Chun, **Interaction between host cell proteins and open reading frames of porcine circovirus type 2**, *Journal of Animal Science and Technology* 65 (4), 698, 2023
- SP Choi, SW Park, SJ Kang, SK Lim, MS Kwon, HJ Choi, T Chun, **Monitoring mRNA expression patterns in macrophages in response to two different strains of probiotics**, *Food Science of Animal Resources* 43 (4), 703, 2023
- CY Choi, CH Lee, J Yang, SJ Kang, IB Park, SW Park, NY Lee, HB Hwang, et al., **Efficacies of potential probiotic candidates isolated from traditional fermented Korean foods in stimulating immunoglobulin A**

secretion, *Food Science of Animal Resources* 43 (2), 346, 2023

- **SJ Kang**, J Yang, NY Lee, CH Lee, IB Park, SW Park, HJ Lee, HW Park, et al., **Monitoring cellular immune responses after consumption of selected probiotics in immunocompromised mice**, *Food Science of Animal Resources* 42 (5), 903, 2022
- **SJ Kang**, IB Park, T Chun, **Open reading frame 5 protein of porcine circovirus type 2 induces RNF128 (GRAIL) which inhibits mRNA transcription of interferon- β in porcine epithelial cells**, *Research in Veterinary Science* 140, 79-82, 2021
- **SJ Kang**, T Chun, **Structural heterogeneity of the mammalian polycomb repressor complex in immune regulation**, *Experimental & Molecular Medicine* 52 (7), 1004-1015, 2020
- SP Choi, YC Choi, J Yang, CY Choi, CH Lee, **SJ Kang**, IB Park, T Chun, **Monitoring mRNA transcription of genes involved in early pregnancy from endometrium and peripheral blood mononuclear cells of pregnant pigs with different parity**, *Reproduction in Domestic Animals* 53 (6), 1594-1599, 2018
- CY Choi, YC Choi, IB Park, CH Lee, **SJ Kang**, T Chun, **The ORF5 protein of porcine circovirus type 2 enhances viral replication by dampening type I interferon expression in porcine epithelial cells**, *Veterinary Microbiology* 226, 50-58, 2018

Patents

Registration

- T Chun, J Yang, CH Lee, **SJ Kang**, IB Park, **Chimeric antigen receptor specifically binding to CD38 and use thereof**, *Korean Intellectual Property Office*, 10-2021-0025895, October 2021
- T Chun, J Yang, CH Lee, **SJ Kang**, IB Park, **Chimeric antigen receptor specifically binding to VLA-4 and use thereof**, *Korean Intellectual Property Office*, 10-2021-0025905, March 2021

Application

- **SJ Kang**, H Shin, **Method and system for classifying intrinsically disordered proteins based on amino acid sequences using quantum neural networks**, *Korean Intellectual Property Office*, 10-2025-0078871, June 2025
- T Chun, IB Park, **SJ Kang**, HJ Lee, **Regulation of BCL11A expression by LDB1 transcriptional regulator and uses thereof**, *Korean Intellectual Property Office*, 10-2025-0010845, January 2025
- T Chun, IB Park, **SJ Kang**, HJ Lee, **Regulation of PLZF expression by LDB1 transcriptional regulator and uses thereof**, *Korean Intellectual Property Office*, 10-2025-0010844, January 2025
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Chimeric antigen receptor specifically binding to CD138 and use thereof**, *Korean Intellectual Property Office*, 10-2023-0058232, May 2023
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Recombinant protein which recognizes CADM1 and the composition comprising the same for treating cancer**, *Korean Intellectual Property Office*, 10-2023-0061710, May 2023
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Chimeric antigen receptor comprising BAFF extracellular domain and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0153707, November 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Chimeric antigen receptor specifically binding to CD47 and use thereof**, *Korean Intellectual Property Office*, 10-2022-0138909, October 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Transformed professional antigen presenting cells specifically binding to antigen containing chimeric antigen receptor (CAR) and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0138790, October 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Transformed professional antigen presenting cells specifically binding to antigen containing chimeric antigen receptor (CAR) and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0136876, October 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Chimeric antigen receptor specifically binding to RANK ligand and use thereof**, *Korean Intellectual Property Office*, 10-2022-0124839, September 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Recombinant protein which**

induces cytotoxicity in CD30 expressing cell and the composition comprising the same for treating cancer, *Korean Intellectual Property Office*, 10-2022-0124005, September 2022

- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Chimeric antigen receptor specifically binding to HLA class II and use thereof**, *Korean Intellectual Property Office*, 10-2022-0122831, September 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Transformed professional antigen presenting cells specifically binding to antigen containing chimeric antigen receptor (CAR) and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0131801, September 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Transformed professional antigen presenting cells specifically binding to antigen containing chimeric antigen receptor (CAR) and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0132401, September 2022
- HS Jin, T Chun, J Yang, CH Lee, NY Lee, IB Park, **SJ Kang**, SW Park, HJ Lee, **Transformed professional antigen presenting cells specifically binding to antigen containing chimeric antigen receptor (CAR) and uses thereof**, *Korean Intellectual Property Office*, 10-2022-0150814, November 2021

Research Projects

1. **1465039780** 04/2022 – 12/2023
Title: Development of universal SARS-CoV-2 and sarbecovirus vaccine candidate using NDV vector platform and evaluation of broad-spectrum protective efficacy
Source: Korea Health Industry Development Institute
Role: Research Assistant (PI: Taehoon Chun)
2. **1475013349** 02/2022 – 12/2023
Title: Development of mandatory vaccine evaluation method using new platform(2)
Source: Ministry of Food and Drug Safety
Role: Research Assistant (PI: Taehoon Chun)
3. **PJ013271012018** 03/2018 – 12/2020
Title: Development of genetic markers associated to the resistance to PMWS by dissecting the molecular mechanism of viral host interaction
Source: Rural Development Administration
Role: Research Assistant (PI: Taehoon Chun)

Conference Presentations

Poster Presentations

- **SJ Kang**, T Chun, **Immune regulation of mammalian polycomb repressor and the relationship between its structural heterogeneity**, *Korean Association of Immunologists (KAI)*, Seoul, South Korea, 2020
- **SJ Kang**, T Chun, **Administration of Lactobacillus Plantarum CJLP133 and CJLP243 Ameliorates the Symptoms of Allergic Rhinitis in BP-Inhaled Mouse Model**, *Korean Association of Immunologists (KAI)*, Seoul, South Korea, 2019
- **SJ Kang**, T Chun, **Microorganisms from Kimchi, GK3 and Baechu10, Change the Cytokine Production Pattern of Host Peritoneal Macrophages**, *Korean Association of Immunologists (KAI)*, Seoul, South Korea, 2018

Books

- **SJ Kang**, **2026 Science Trends: Multi-omics**, *Kindle*, ISBN 9798266588004, September 2025
- **SJ Kang**, **2026 Life Science Trends: Multi-omics (Korean Edition)**, *Purple*, ISBN 9788924169034, August 2025
- **SJ Kang**, **2026 Science Trends: Personalized Medicine (Korean Edition)**, *Purple*, ISBN 9788924169041, August 2025
- **SJ Kang**, **2025 Science Trends: Artificial Intelligence**, *Kindle*, ISBN 979-8307079270, January 2025
- **SJ Kang**, **2025 Science Trends: Artificial Intelligence (Korean Edition)**, *Purple*, ISBN 9788924143621, December 2024

Certifications

- **Big Data Analysis Engineer**, *Korea Data Agency*, 2025, BAE-010002862
- **ADsP (Advanced Data Analytics Semi-Professional)**, *Korea Data Agency*, 2024, ADsP-040004911

Technologies

Programming Languages: Python, R

Data Science & Bioinformatics: Machine Learning, Deep Learning, Quantum Neural Networks, Multi-omics Analysis (Genomics, Transcriptomics, Proteomics), Computational Biology, Statistical Analysis, Sequence Analysis, Structural Bioinformatics

Bioinformatics Tools: PyTorch, TensorFlow, scikit-learn, Biopython, RStudio, Pandas, NumPy

Molecular Biology Techniques: CRISPR/Cas9 Gene Editing, CAR-T Cell Therapy, Protein Design & Engineering, Protein Expression & Production, Protein Purification, Molecular Cloning, Cell Culture, Virology, Immunology Assays, Xenotransplantation

Laboratory Skills: Western Blot, PCR/qPCR, Flow Cytometry, ELISA, Chromatography, Electrophoresis, SPR (Surface Plasmon Resonance), Protein-Protein Interaction Analysis, K_d Measurement, Cell-based Assays, Animal Models (Mouse, Pig)

Research Interests

Computational Biology, Machine Learning in Drug Discovery, Protein Engineering, CAR-T Cell Therapy, Multi-omics Integration, Quantum Computing in Biology